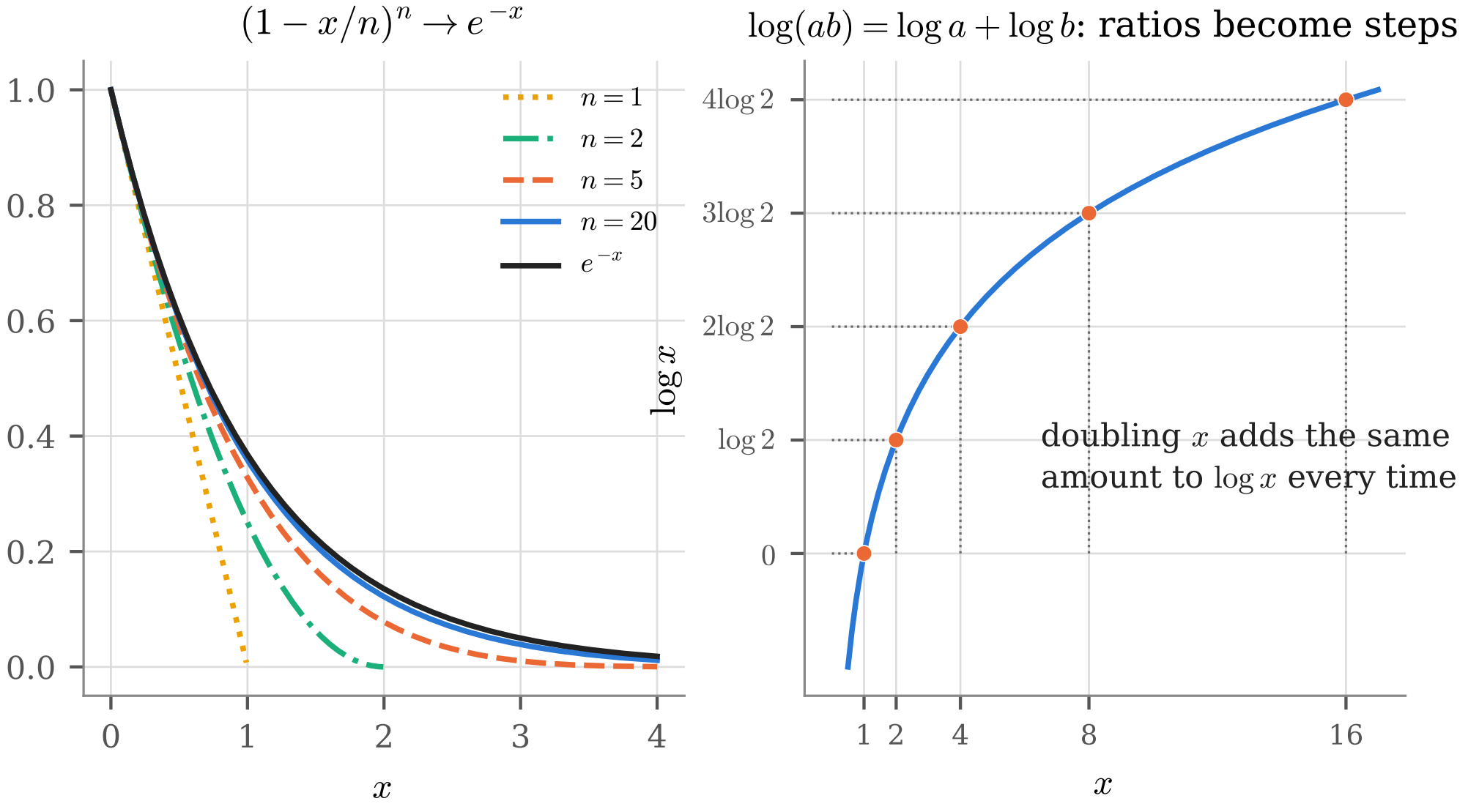
$(1 - x/n)^n \to e^{-x}$
$n = 1$
$n = 2$
$n = 5$
$n = 20$
e^{-x}
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
x
$\log(ab) = \log a + \log b$: ratios become steps
4log 2
3log 2
2log 2
log 2
0
log x
doubling x adds the same
amount to log x every time
1
2
4
8
16
x